

Rounding switching controls under a hard switch budget: sharp minimax bounds and exact algorithms

Abstract

Combinatorial integral approximation replaces a relaxed mixture of control modes by one active mode at a time while controlling cumulative error. A hard bound on the number of switches changes the approximation problem: even on arbitrarily fine time grids, the worst-case error can remain proportional to the horizon. We study this error as a function of the number of modes, the switch budget, and the time grid. We establish exact uniform-input formulas and, for at least three modes, the full continuous one-switch minimax value, with matching extremizers and constructive proofs for measurable relaxed controls. These results give explicit counterexamples to a published switching-budget conjecture. The continuous and grid problems are formulated separately, so that bounds for individual inputs and worst-case bounds have unambiguous meanings.

1 Problem and basic properties

A relaxed switching control allocates one unit of activity among finitely many modes. Rounding selects one mode at a time. The cumulative allocation error is the objective of combinatorial integral approximation (CIA), the rounding subproblem in a decomposition of mixed-integer optimal control. A restriction on the number of mode changes is a hard constraint on that subproblem. Sager and Zeile [1] study this formulation, develop maximum dwell rounding, and give sharp bounds and a conjecture for its worst-case error. Our focus is the extremal rounding problem itself. In particular, a bound on cumulative allocation error is not an assertion that the same schedule optimizes an underlying nonlinear control objective.

1.1 Controls, schedules, and the two error criteria

Fix a horizon $T > 0$ and an integer number of modes $n \geq 2$. Throughout, switch budgets s are nonnegative integers, activation-block counts k are positive integers, and grid sizes N are positive integers. Write $[n] = \{1, \dots, n\}$ and

$$\Delta_n = \left\{ a \in \mathbb{R}^n : a_i \geq 0, \sum_{i=1}^n a_i = 1 \right\}.$$

A relaxed control is a Lebesgue-measurable map $\alpha : [0, T] \rightarrow \Delta_n$, considered up to equality almost everywhere. Its cumulative allocation and terminal masses are

$$A_i(t) = \int_0^t \alpha_i(u) du, \quad m_i = A_i(T). \quad (1)$$

Thus A_i is nondecreasing and 1-Lipschitz, $A_i(0) = 0$, and $\sum_i A_i(t) = t$. Conversely, every vector of functions with these properties is the cumulative allocation of such a control:

Lipschitz functions are absolutely continuous and differentiable almost everywhere, their derivatives are nonnegative, and differentiation of the sum gives a simplex-valued vector almost everywhere. Denote this class of cumulative allocations by $\mathcal{A}_n(T)$. We use either α or A to specify an input.

An integer control, or schedule, selects a coordinate vector e_i on each of finitely many intervals. Its initial activation is free: only changes between two successive active modes count as switches. Its values at isolated times are immaterial. Let $\mathcal{W}_{n,k}(T)$ be the set of cumulative occupations

$$W_i(t) = \int_0^t \omega_i(u) du$$

of schedules with at most k positive-length activation blocks, equivalently at most $s = k - 1$ switches. Unless explicitly stated otherwise, repeated modes are allowed, the initial and final modes are unrestricted, and there are no dwell-time or transition restrictions. A parameterization of this entire set uses a word $p \in [n]^k$ and ordered times

$$0 = t_0 \leq t_1 \leq \dots \leq t_k = T, \quad \omega(t) = e_{p_j} \quad (t_{j-1} < t < t_j). \quad (2)$$

Zero-length intervals can be deleted and consecutive equal modes can be merged. These operations cannot increase the switch count. Consequently this parameterization includes shorter schedules without imposing distinctness on competing modes.

The full error and the one-sided error are respectively

$$D(A, W) = \max_{i \in [n]} \max_{0 \leq t \leq T} |A_i(t) - W_i(t)|, \quad D^-(A, W) = \max_{i \in [n]} \max_{0 \leq t \leq T} (W_i(t) - A_i(t)). \quad (3)$$

The minus superscript refers to the negative part of the discrepancy $A - W$; the displayed one-sided maximum is nonnegative because it includes $t = 0$. For a fixed input, define the continuous instance optima

$$\text{OPT}_s(A) = \min_{W \in \mathcal{W}_{n,s+1}(T)} D(A, W), \quad \text{OPT}_k^-(A) = \min_{W \in \mathcal{W}_{n,k}(T)} D^-(A, W). \quad (4)$$

The corresponding minimax values are

$$F_{n,s}(T) = \sup_{A \in \mathcal{A}_n(T)} \text{OPT}_s(A), \quad G_{n,k}^-(T) = \sup_{A \in \mathcal{A}_n(T)} \text{OPT}_k^-(A). \quad (5)$$

The indexing is intentional: F counts switches, whereas G^- counts activation blocks, with $k = s + 1$. Both infima in (4) and both suprema in (5) are attained, as shown below.

1.2 Fixed grids and endpoint evaluation

A grid is a strictly increasing sequence $\mathcal{T} = (x_0, \dots, x_N)$ with $x_0 = 0$, $x_N = T$. Its cell lengths are $\Delta_j = x_j - x_{j-1}$ and $\bar{\Delta} = \max_j \Delta_j$. Let $\mathcal{A}_n(\mathcal{T})$ consist of cumulative allocations of controls constant on each cell, and let $\mathcal{W}_{n,k}(\mathcal{T})$ restrict the switching times to grid points. Set

$$\text{OPT}_s^{\mathcal{T}}(A) = \min_{W \in \mathcal{W}_{n,s+1}(\mathcal{T})} D(A, W), \quad F_{n,s}^{\mathcal{T}} = \max_{A \in \mathcal{A}_n(\mathcal{T})} \text{OPT}_s^{\mathcal{T}}(A). \quad (6)$$

The instance optimum is defined even for $A \in \mathcal{A}_n(T)$ that is not piecewise affine. The grid minimax in (6) does restrict the input to cellwise constant controls. Analogously, use $\text{OPT}_k^{-,\mathcal{T}}$ and $G_{n,k}^{-,\mathcal{T}}$ for the one-sided criterion. On the unit grid $(0, 1, \dots, N)$ we abbreviate the full minimax by $F_{n,s}^{\text{unit}}(N)$. Every grid with cell length Δ is obtained from a unit grid by scaling all errors by Δ .

For the same input, $\text{OPT}_s(A) \leq \text{OPT}_s^{\mathcal{T}}(A)$. This fact alone does not compare the two full minimax values, whose input classes differ. The following observation permits exact evaluation even for merely measurable inputs and will be used throughout.

Lemma 1.1 (Endpoint monotonicity). *On any interval on which the selected mode is p , the discrepancy $A_p - W_p$ is nonincreasing and every other discrepancy $A_i - W_i$ is nondecreasing. Hence both errors in (3) are determined by the activation-block endpoints. In particular, for a grid schedule they are determined by grid endpoints, even when α varies inside grid cells.*

Proof. The absolutely continuous discrepancies have derivatives $\alpha_p - 1 \leq 0$ and $\alpha_i \geq 0$ for $i \neq p$, almost everywhere on the interval. Integrating these inequalities proves monotonicity. A monotone real function takes its largest absolute value and each of its one-sided extrema at an endpoint. $\square$

Replacing a measurable control by its cell averages preserves every grid endpoint $A_i(x_j)$. Lemma 1.1 therefore shows that this replacement preserves the error of every grid schedule, and hence its grid instance optimum. This observation does not say that averaging preserves its continuous instance optimum. It does, however, give the minimax comparison

$$F_{n,s}(T) \leq F_{n,s}^T, \quad G_{n,k}^-(T) \leq G_{n,k}^{-,T}. \quad (7)$$

Indeed, for any measurable input A , its cell average $\bar{A}$ satisfies $\text{OPT}_s(A) \leq \text{OPT}_s^T(A) = \text{OPT}_s^T(\bar{A}) \leq F_{n,s}^T$. Taking the supremum proves the first inequality; the one-sided argument is identical.

1.3 Existence, continuity, and scaling

Equip vector-valued cumulative functions with $\|A\|_\infty = \max_i \max_{t \in [0,T]} |A_i(t)|$.

Proposition 1.2 (Attainment and stability). *The sets $\mathcal{A}_n(T)$ and $\mathcal{W}_{n,k}(T)$ are compact in the uniform norm. Each of the four instance optimum maps defined above is 1-Lipschitz in A . All continuous and grid instance optima and minimax values are attained.*

Proof. The cumulative allocations are uniformly bounded and equicontinuous. Their uniform limits preserve monotonicity, the Lipschitz constants, initial values, and the sum identity, and therefore belong to $\mathcal{A}_n(T)$ by the converse following (1). The Arzelà–Ascoli theorem gives compactness.

For a fixed word p , the ordered time simplex in (2) is compact. Its cumulative occupations are continuous in those times. Indeed, the useful representation

$$W_i(t) = \mathbf{1}_{\{p_k=i\}}t + \sum_{j=1}^{k-1} (\mathbf{1}_{\{p_j=i\}} - \mathbf{1}_{\{p_{j+1}=i\}}) \min\{t, t_j\}$$

gives $\|W(\cdot; t_1, \dots, t_{k-1}) - W(\cdot; t'_1, \dots, t'_{k-1})\|_\infty \leq \sum_{j=1}^{k-1} |t_j - t'_j|$. Taking the finite union over words proves compactness of $\mathcal{W}_{n,k}(T)$. Both error criteria are continuous, so their minima are attained. For either criterion d , the inequality $|d(A, W) - d(B, W)| \leq \|A - B\|_\infty$ holds for every W . Taking minima shows the stated Lipschitz property. Compactness of $\mathcal{A}_n(T)$ then gives attainment of the outer maxima. For grids, there are finitely many schedules and the input is a product of N compact simplexes, proving the analogous claims. $\square$

Changing time by $t = Tu$ is a bijection of both input and schedule classes and multiplies errors by T . Thus

$$F_{n,s}(T) = TF_{n,s}(1), \quad G_{n,k}^-(T) = TG_{n,k}^-(1). \quad (8)$$

The same statement applies to instance optima and a simultaneously rescaled grid. At $T = 0$ all errors are zero; results stated for $T > 0$ extend with this convention. Enlarging the switch budget cannot increase any instance optimum or minimax value.

Proposition 1.3 (No-switch baseline). *For every $n \geq 2$, $F_{n,0}(T) = T(1 - 1/n)$. On any grid, $F_{n,0}^T = T(1 - 1/n)$ as well.*

Proof. Activating mode p throughout gives error $\max\{T - m_p, \max_{i \neq p} m_i\} = T - m_p$, since $\sum_{i \neq p} m_i = T - m_p$. Choosing a largest terminal mass gives error at most $T(1 - 1/n)$. The uniform control $\alpha_i = 1/n$ gives equality and belongs to every grid input class. $\square$

2 Uniform inputs and the exact one-switch minimax

We first solve a benchmark family for every budget with fewer activation blocks than modes. This family gives a dimension-dependent obstruction to small error. We then prove that its obstruction, together with an omitted-mode obstruction, determines the full continuous one-switch minimax for at least three modes.

2.1 Exact uniform-input error in continuous time

Write $A_i^{\text{unif}}(t) = t/n$ and let $E_{n,s}(T) = \text{OPT}_s(A^{\text{unif}})$.

Theorem 2.1 (Uniform controls). *Let $n \geq 2$ and $0 \leq s \leq n - 2$. With $k = s + 1$ and $r = n/(n - 1)$,*

$$E_{n,s}(T) = T \max \left\{ \frac{1}{n}, \frac{1}{n(r^k - 1)} \right\}. \quad (9)$$

A schedule using k distinct modes attains this value, although competing schedules may repeat modes.

Proof. Consider any schedule with $m \leq k < n$ positive-length blocks and error E . At least one mode is unused, so $E \geq T/n$. Let $0 = t_0 < \dots < t_m = T$ be its block endpoints. At the end of block j , the occupation of its active mode p_j is at least $t_j - t_{j-1}$, even if p_j occurred earlier. The inequality $W_{p_j}(t_j) - t_j/n \leq E$ implies

$$t_j \leq r t_{j-1} + r E. \quad (10)$$

Iterating from zero yields $T \leq nE(r^m - 1) \leq nE(r^k - 1)$ and proves both lower bounds.

For attainment, set

$$E_0 = \frac{T}{n(r^k - 1)}, \quad t_j = T \frac{r^j - 1}{r^k - 1} \quad (0 \leq j \leq k),$$

and choose a distinct mode on each block. Before its activation, the selected mode has discrepancy t/n ; during activation its discrepancy decreases; after activation it increases again. At its block endpoint,

$$A_{p_j}(t_j) - W_{p_j}(t_j) = \frac{t_j}{n} - (t_j - t_{j-1}) = -E_0.$$

Every positive discrepancy is at most T/n , and no selected mode's negative discrepancy exceeds E_0 . Unused modes also have error at most T/n . This proves attainment of (9). $\square$

For one switch, this gives

$$E_{n,1}(T) = T \max \left\{ \frac{1}{n}, \frac{(n-1)^2}{n(2n-1)} \right\} \quad (n \geq 3). \quad (11)$$

For every fixed $s \geq 0$, the uniform value tends to $T/(s+1)$ as $n \rightarrow \infty$. Indeed, with fixed $k = s+1$, $(1 + 1/(n-1))^k - 1 = k/(n-1) + O_k(n^{-2})$, so $n(r^k - 1) \rightarrow k$. This is an obstruction supplied by one input family; it does not presume that this family maximizes the error over all inputs.

2.2 A scalar recurrence on uniform grids

Let $E_{n,s}^{\text{unit}}(N)$ be the grid instance optimum for the uniform control on N unit cells.

Theorem 2.2 (Exact integer recurrence). *Under $n \geq 2$, $0 \leq s \leq n-2$, and $N \geq 1$, define, for integers $K \geq N$,*

$$b_0(K) = 0, \quad b_{j+1}(K) = \left\lfloor \frac{nb_j(K) + K}{n-1} \right\rfloor. \quad (12)$$

If K_ is the least such integer for which $b_{s+1}(K_*) \geq N$, then $E_{n,s}^{\text{unit}}(N) = K_*/n$. Moreover,*

$$E_{n,s}(N) \leq E_{n,s}^{\text{unit}}(N) < E_{n,s}(N) + \frac{n-1}{n}. \quad (13)$$

Proof. At integer times, every discrepancy is a multiple of $1/n$; by Lemma 1.1 an optimal error is therefore K/n for an integer K . Every admissible schedule omits a mode, giving $K \geq N$. The block-end argument (10), with integer block endpoints, gives $t_j \leq \lfloor (nt_{j-1} + K)/(n-1) \rfloor$. Monotonicity of this recurrence implies $t_j \leq b_j(K)$ and hence necessity of $b_{s+1}(K) \geq N$. Fewer blocks can be padded by zero-length blocks; alternatively the nondecreasing iterates give the same conclusion.

Conversely, suppose the condition holds. Put $t_j = \min\{N, b_j(K)\}$ and stop when N is first reached. Since the recurrence cannot leave zero if $b_1(K) = 0$, the assumed condition implies $b_1(K) \geq 1$, or $K \geq n-1$. It follows that the iterates strictly increase before truncation. Assign a distinct mode to each positive-length block. Its block-end inequality gives $W_i(t_j) - t_j/n \leq K/n$. Before and after that block, its discrepancy is nondecreasing, so this bounds its negative part throughout. Positive errors of selected and unused modes are at most $N/n \leq K/n$. There are at most $s+1$ blocks, proving sufficiency.

The predicate is monotone in K , and $K = N(n-1)$ already reaches N in one step. Thus K_* exists and binary search may use $N \leq K \leq N(n-1)$, with $s+1$ recurrence evaluations per trial. To prove the strict gap, write $E = E_{n,s}(N)$ and $C = \lceil nE \rceil$, and test $K = C + n - 2 \geq N$. For every integer b ,

$$\left\lfloor \frac{nb + C + n - 2}{n-1} \right\rfloor = \left\lceil \frac{nb + C}{n-1} \right\rceil \geq rb + rE.$$

The discrete recurrence therefore dominates the continuous recurrence starting at zero, which reaches at least N in $s+1$ steps by (9). Consequently $E_{n,s}^{\text{unit}}(N) \leq K/n < E + (n-1)/n$. The lower inequality follows by restricting continuous switching times to integer times for the same input. $\square$

2.3 The continuous one-switch minimax

For two distinct modes p, q , let $W^{p,q,\tau}$ activate p before time τ and q afterward, allowing $\tau = 0$ or T . These endpoints include constant schedules.

Lemma 2.3 (The three-term formula). *For $n \geq 2$ and $0 \leq \tau \leq T$, taking the maximum over omitted terminal masses as zero when $n = 2$,*

$$D(A, W^{p,q,\tau}) = \max \left\{ \max_{i \notin \{p,q\}} m_i, \quad \tau - A_p(\tau), \quad T - m_q - \tau \right\}. \quad (14)$$

Proof. The omitted modes increase to their terminal masses. The discrepancy of p decreases from zero to $A_p(\tau) - \tau$ and then increases to $m_p - \tau$. Its possible positive final error is

bounded by $T - m_q - \tau$, because $m_p + m_q \leq T$. The discrepancy of q increases to $A_q(\tau)$ and then decreases to $m_q - (T - \tau)$. Its positive error at activation is at most $\tau - A_p(\tau)$, because $A_p(\tau) + A_q(\tau) \leq \tau$. These comparisons bound every endpoint extremum by the right-hand side. Conversely, the first two displayed terms are actual errors. If the last term is positive it is an actual negative terminal discrepancy; if it is nonpositive it is dominated by the nonnegative second term. This proves the equality, including $\tau = 0, T$. $\square$

Theorem 2.4 (Exact continuous one-switch minimax). *For every $n \geq 3$,*

$$F_{n,1}(T) = H_n(T) := T \max \left\{ \frac{1}{3}, \frac{(n-1)^2}{n(2n-1)} \right\}. \quad (15)$$

For $n = 3, 4$, three consecutive distinct pure modes, each of duration $T/3$, attain the maximum. For $n \geq 5$, the uniform control attains it. A schedule with the asserted error bound can be constructed using terminal masses and cumulative allocations at at most two additional times.

Proof. By scaling take $T = 1$ and put $E = H_n(1)$. Directly from the formula, $1/3 \leq E < 1/2$.

First suppose some $m_q \geq E$. At most one other mode has mass strictly larger than E : two such modes together with q would have total exceeding $3E \geq 1$. Choose $p \neq q$ to include that mode if it exists, and choose any other mode otherwise. All omitted masses are at most E . With $\tau = \max\{0, 1 - m_q - E\}$ we have $0 \leq \tau \leq \max\{0, 1 - 2E\} \leq E$. Thus $\tau - A_p(\tau) \leq E$ and $1 - m_q - \tau \leq E$, and Lemma 2.3 gives a feasible schedule.

It remains to treat the case $m_i < E$ for every mode. Let q index a largest mass and r a second largest, breaking ties arbitrarily. Define

$$t_q = 1 - m_q - E, \quad t_r = 1 - m_r - E, \quad h_i = 1 - m_i - 2E.$$

Since $m_i < E < 1/2$, the times satisfy $0 < t_q \leq t_r < 1$. Consider all schedules $p \rightarrow q$, $p \neq q$, at time t_q , and also $q \rightarrow r$ at time t_r . Their omitted masses are below E , and their last terms in (14) equal E . If all these schedules had error strictly larger than E , the middle terms would force

$$A_p(t_q) < h_q \quad (p \neq q), \quad A_q(t_q) \leq A_q(t_r) < h_r.$$

Summing gives $t_q < (n-1)h_q + h_r$, or

$$(2n-1)E < n-1 - (n-2)m_q - m_r. \quad (16)$$

But $m_q \geq 1/n$ and $m_r \geq (1 - m_q)/(n-1)$. For $n \geq 3$ the coefficient $n-2-1/(n-1)$ is positive, so

$$(n-2)m_q + m_r \geq \left(n-2 - \frac{1}{n-1}\right)m_q + \frac{1}{n-1} \geq \frac{n-1}{n}.$$

Equation (16) would imply $(2n-1)E < (n-1)^2/n$, contrary to the definition of E . At least one candidate therefore has error at most E .

For a matching lower bound of $1/3$, use three consecutive distinct pure modes of duration $1/3$. Every schedule with at most two blocks omits at least one of these modes and incurs its terminal mass. The uniform-input value in (11) supplies the other lower bound. Finally,

$$\frac{(n-1)^2}{n(2n-1)} - \frac{1}{3} = \frac{n^2 - 5n + 3}{3n(2n-1)}$$

is negative at $n = 3, 4$ and positive for every integer $n \geq 5$, giving the claimed transition and extremizers. $\square$

For an arbitrary horizon $T > 0$, set $E = H_n(T)$ and $t_i = T - m_i - E$. The construction requires no search over switching times. Let q and r index a largest and second-largest terminal mass. If $m_q \geq E$, select $r \rightarrow q$ at $\max\{0, T - m_q - E\}$. Otherwise evaluate $A(t_q)$ and choose $p \neq q$ maximizing $A_p(t_q)$. If its candidate fails, the $q \rightarrow r$ candidate at t_r succeeds. Apart from cumulative integration, this requires $O(n)$ arithmetic operations and comparisons. The proof covers ties, zero terminal masses, and switching at an endpoint.

Corollary 2.5 (One-switch grid upper bound). *For every grid $\mathcal{T}$ and every measurable relaxed input,*

$$\text{OPT}_1^{\mathcal{T}}(A) \leq H_n(T) + \frac{\bar{\Delta}}{2} \quad (n \geq 3).$$

The coefficient $H_n(T)/T$ cannot be decreased in a bound whose grid term vanishes under refinement, uniformly over grids and inputs.

Proof. Choose the schedule from Theorem 2.4 and move its switch to a closest grid point. The displacement is at most $\bar{\Delta}/2$. The cumulative occupation of either affected mode changes by at most that displacement, so the triangle inequality gives the bound. No new switch is created. For sharpness of the leading coefficient, use the uniform extremizer when $n \geq 5$. When $n = 3, 4$, use the three pure blocks on successively refined grids containing $T/3$ and $2T/3$. Each input belongs to its grid class, and its grid optimum is at least its continuous optimum $H_n(T)$. $\square$

2.4 Exact three-cell minimax and a published conjecture

Proposition 2.6 (Three unit cells). *For $n \geq 3$, $F_{n,1}^{\text{unit}}(3) = 2 - 3/n$.*

Proof. For arbitrary relaxed input choose q with a largest terminal mass and p with a second largest, and use the three-cell word (p, q, q) . Every omitted mass is at most 1, because the masses sum to 3. The first-cell error is at most 1. At time 2, the selected modes have occupation 1 and cumulative allocations in $[0, 2]$, hence errors at most 1; omitted modes still have errors at most 1. At time 3, $m_p \leq 3/2$ and its occupation is 1, so its error is at most 1. The final mode satisfies $3/n \leq m_q \leq 3$ and has occupation 2, giving error at most $\max\{1, 2 - 3/n\} = 2 - 3/n$. Endpoint monotonicity proves the upper bound. For uniform input, every schedule with at most one switch gives some mode at least two cells. Its negative terminal discrepancy is at least $2 - 3/n$. Thus the upper bound is attained. $\square$

Sager and Zeile [1, Conjecture 1, equation (7.6)] propose, for $n > 2$ and $1 \leq s \leq N - 2$, the worst-case grid value

$$\begin{cases} T/(s+2) + \bar{\Delta}/2, & s \leq n-2, \\ T/(2s+4-n) + \bar{\Delta}/2, & s > n-2. \end{cases} \quad (17)$$

Their initial activation is free, as in our formulation [1, Definition 10]. The following examples satisfy all of the stated restrictions and disprove the first branch even as an upper bound. No conclusion about every case of its second branch is needed here.

Example 2.7 (Five modes and one switch). Take $n = 5$, $T = N = 45$ unit cells, and $\alpha_i = 1/5$ throughout. For any distinct pair of modes and an integer switch time τ , the exact error from (14) is

$$D(\tau) = \max\{9, 4\tau/5, 36 - \tau\}.$$

The two nonconstant terms intersect at $\tau = 20$, with value 16. For $\tau \leq 20$ the last term is at least 16; for $\tau \geq 20$ the middle term is at least 16. Constant schedules are included by

the endpoint choices. Thus the grid optimum is 16, whereas (17) gives $45/3 + 1/2 = 31/2$. On $45m$ unit cells the same construction has optimum $16m$, exceeding the conjectured $15m + 1/2$ for every integer $m \geq 1$. Equivalently, on a fixed horizon T with $45m$ equal cells its optimum is $16T/45$, while the conjectured bound tends to $T/3$. The failure persists under refinement.

Example 2.8 (Smallest admissible number of cells). On three unit cells and with seven uniform modes, the exact grid error is $11/7$ by Proposition 2.6. It exceeds the conjectured $3/3 + 1/2 = 3/2$. The restriction $1 \leq s \leq N - 2$ excludes grids with fewer than three cells, so this witness uses the smallest admissible number of cells.

More generally, for every fixed $s \geq 1$, Theorem 2.1 approaches $T/(s+1) > T/(s+2)$ as the mode count grows. For sufficiently many modes, its continuous uniform value already exceeds the first branch’s proposed leading term. A sufficiently fine uniform grid then makes $\bar{\Delta}/2$ smaller than this strict excess; restricting the uniform input to grid schedules cannot reduce its optimum. Thus this obstruction occurs for every fixed positive switch budget, not only for one switch. In the one-switch case, Theorem 2.4 and Corollary 2.5 give an exact continuous replacement and a valid grid upper bound. The latter is not asserted to be an exact finite-grid equality.

References

- [1] Sebastian Sager and Clemens Zeile. On mixed-integer optimal control with constrained total variation of the integer control. *Computational Optimization and Applications*, 78 (2):575–623, 2021. doi: 10.1007/s10589-020-00244-5. URL <https://doi.org/10.1007/s10589-020-00244-5>.